

Climate Change Governance: Science, Data & Models



Lecture 5: Global Circulation Models (GCMs) and Earth System Models (ESMs)

ENGG562/ENV562 – Fall 2026

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What are Models?

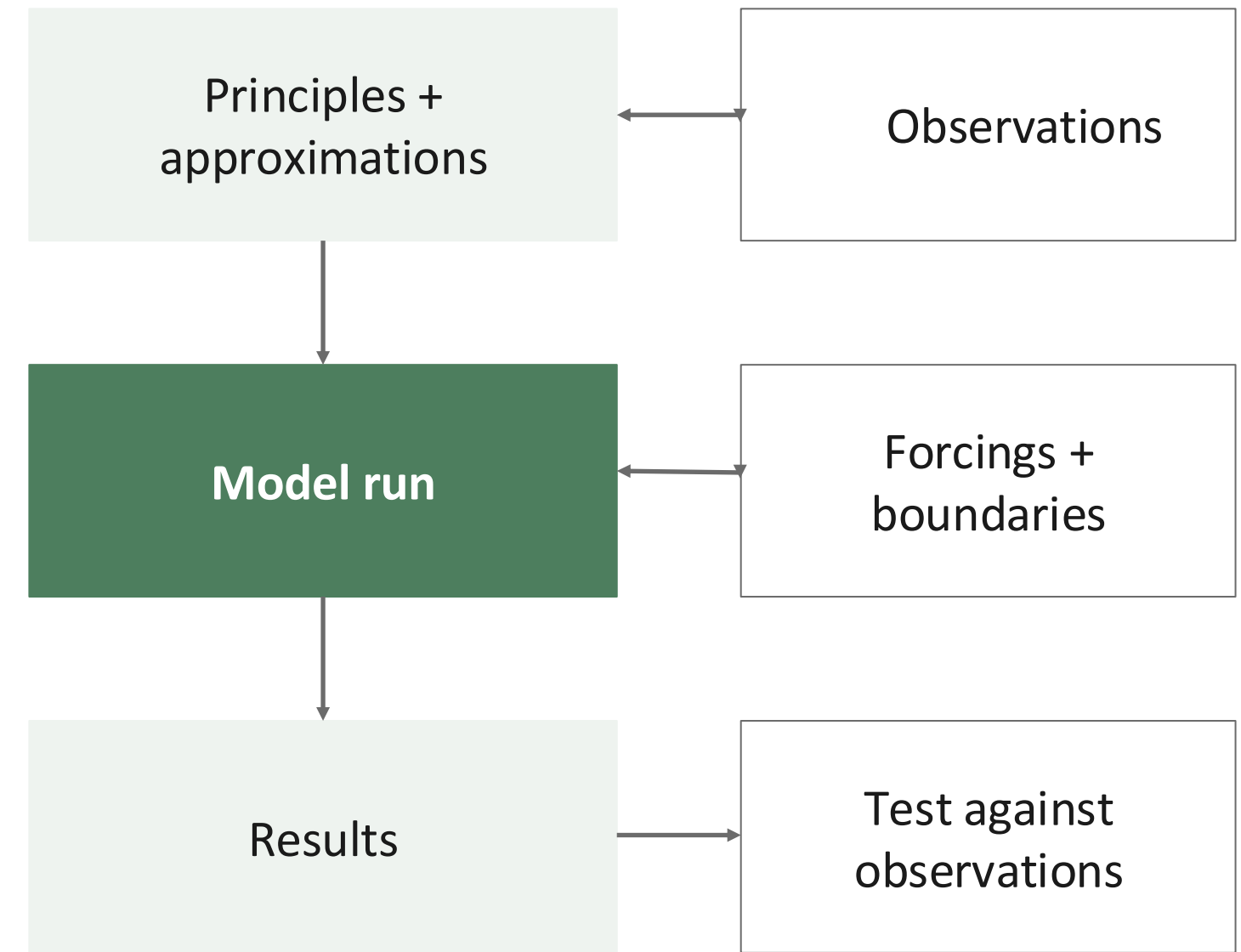
- Models are description of reality but
Models $\neq$ *Reality*
Helps to understand reality

“All models are wrong, but some are useful” – George Box, 1976

Complexity of models can depend on the question you are trying to answer from it.

a climate model could be defined as a mathematical representation of the climate system based on physical, biological and chemical principles

- Built from physical, chemical and biological principles
- Equations too complex to solve by hand, so solved numerically
- Output is discrete: averages over grid boxes and time steps, not points
- Observations enter three times: to build, to force, to test



Development → simulation → evaluation

Climate models are primary tools scientist use to simulate and predict how Earth's climate behaves or will behave over time

Energy Balance Models (EBMs)

- balance incoming solar and outgoing longwave radiation to predict surface temperature, in zero or one dimension.

Radiative-Convective (RC) or Single-Column Models

- resolve one vertical column, combining radiation and convection to produce a temperature profile and lapse rate.

General Circulation Models (GCMs)

- solve fundamental physics equations to simulate atmospheric and ocean circulation.

Earth System Models (ESMs)

- build on GCMs by adding biogeochemical cycles (carbon, nitrogen and phosphorus) and ecosystem dynamics.

Integrated Assessment Models (IAMs)

- couple a simplified climate to models of energy, land use and the economy, to price policy choices rather than to study physics.

A GCM typically contains enough computer code to fill 18,000 pages of printed text

- Conservation of momentum (Newton, on a rotating sphere)

NEWS | MATH

How an AI math breakthrough ignited a controversy

OpenAI spent millions to solve the Navier-Stokes problem, but who should get credit?

8 SEP 2026 • 7:45 PM ET • BY CELINA ZHAO, ADRIAN CHO

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OpenAI

Research Products Business Developers Company Foundation

September 8, 2026 Research Publication

On the Navier–Stokes Millennium Prize Problem

Read the paper Link to Lean formalized proof >

Listen to article

8:43

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Navier–Stokes Equations

3 – dimensional – unsteady

Glenn Research Center

Coordinates: (x,y,z)

Time : t

Pressure: p

Heat Flux: q

Velocity Components: (u,v,w)

Density: ρ

Stress: τ

Reynolds Number: Re

Total Energy: Et

Prandtl Number: Pr

Continuity:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0$$

X – Momentum:

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2)}{\partial x} + \frac{\partial(\rho uv)}{\partial y} + \frac{\partial(\rho uw)}{\partial z} = -\frac{\partial p}{\partial x} + \frac{1}{Re_r} \left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} \right]$$

Y – Momentum:

$$\frac{\partial(\rho v)}{\partial t} + \frac{\partial(\rho uv)}{\partial x} + \frac{\partial(\rho v^2)}{\partial y} + \frac{\partial(\rho vw)}{\partial z} = -\frac{\partial p}{\partial y} + \frac{1}{Re_r} \left[\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} \right]$$

Z – Momentum:

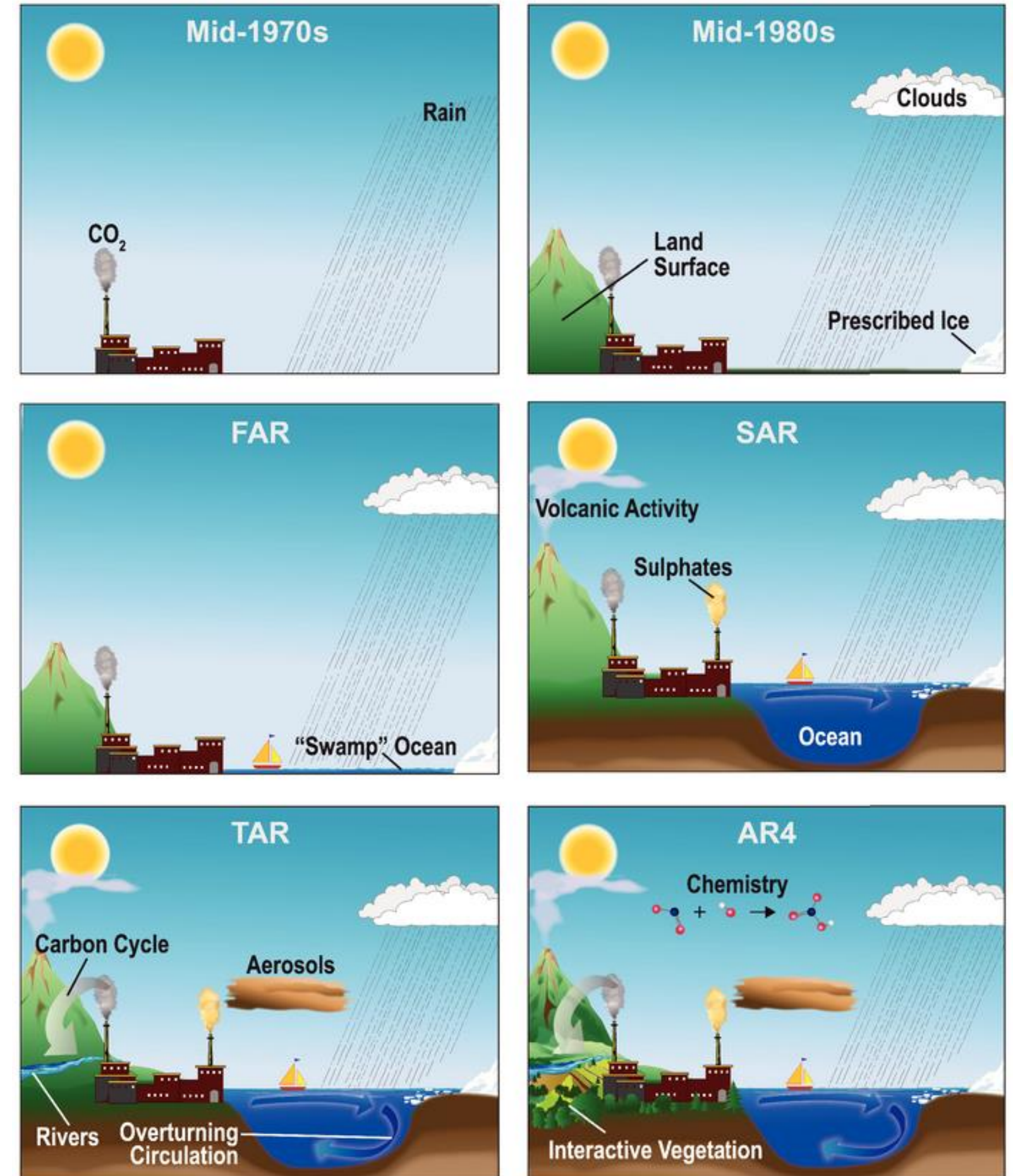
$$\frac{\partial(\rho w)}{\partial t} + \frac{\partial(\rho uw)}{\partial x} + \frac{\partial(\rho vw)}{\partial y} + \frac{\partial(\rho w^2)}{\partial z} = -\frac{\partial p}{\partial z} + \frac{1}{Re_r} \left[\frac{\partial \tau_{xz}}{\partial x} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} \right]$$

$$\frac{\partial(uE_r)}{\partial x} + \frac{\partial(vE_r)}{\partial y} + \frac{\partial(wE_r)}{\partial z} = -\frac{\partial(up)}{\partial x} - \frac{\partial(vp)}{\partial y} - \frac{\partial(wp)}{\partial z} - \frac{1}{Re_r Pr_r} \left[\frac{\partial q_x}{\partial x} + \frac{\partial q_y}{\partial y} + \frac{\partial q_z}{\partial z} \right]$$

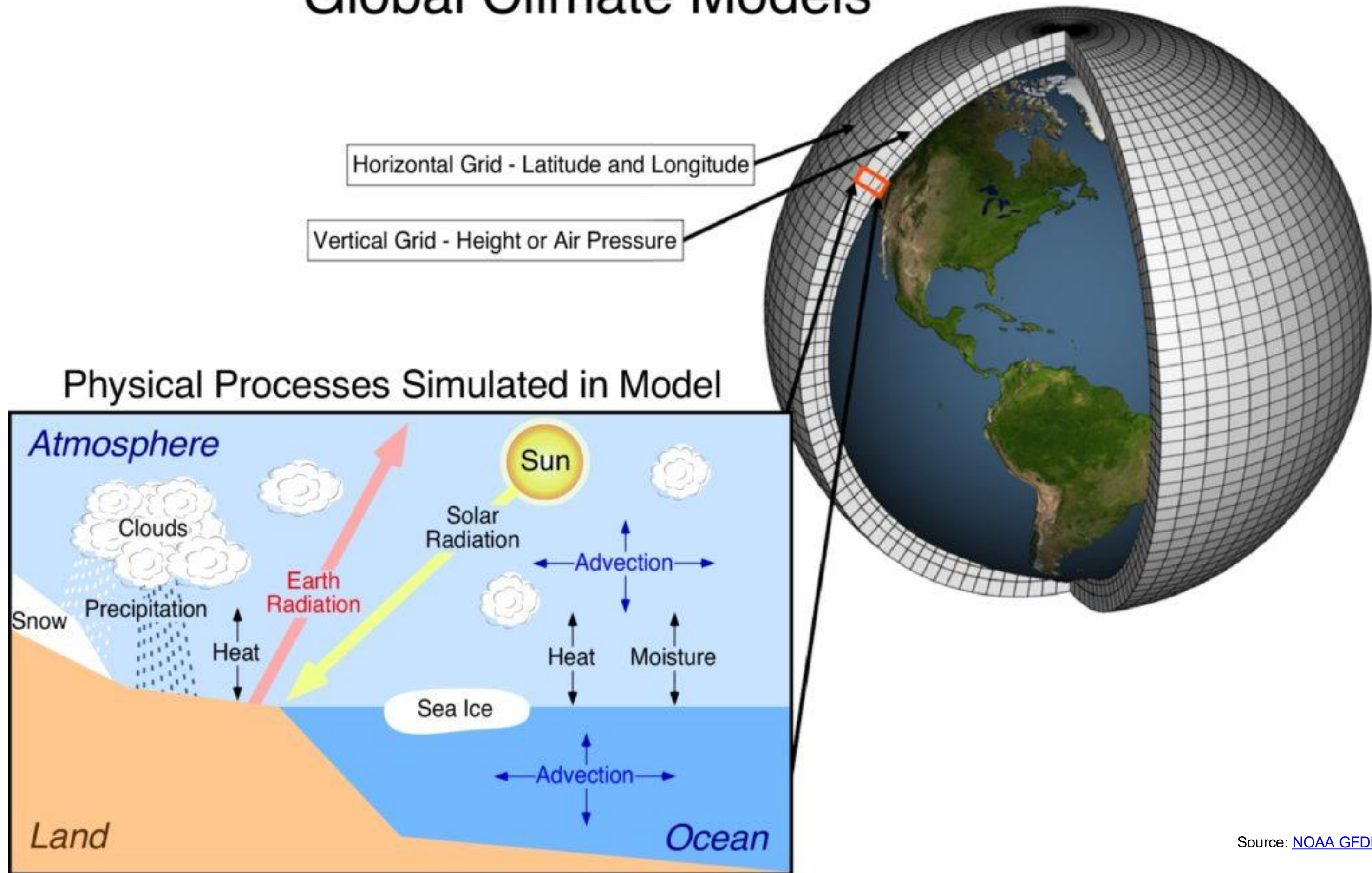
$$\frac{1}{Re_r} \left[\frac{\partial}{\partial x} (u \tau_{xx} + v \tau_{xy} + w \tau_{xz}) + \frac{\partial}{\partial y} (u \tau_{xy} + v \tau_{yy} + w \tau_{yz}) + \frac{\partial}{\partial z} (u \tau_{xz} + v \tau_{yz} + w \tau_{zz}) \right]$$

Illustration of the processes added to global climate models over the decades, from the mid-1970s, through the first four IPCC assessment reports: first (“FAR”) published in 1990, second (“SAR”) in 1995, third (“TAR”) in 2001 and fourth (“AR4”) in 2007.

The World in Global Climate Models



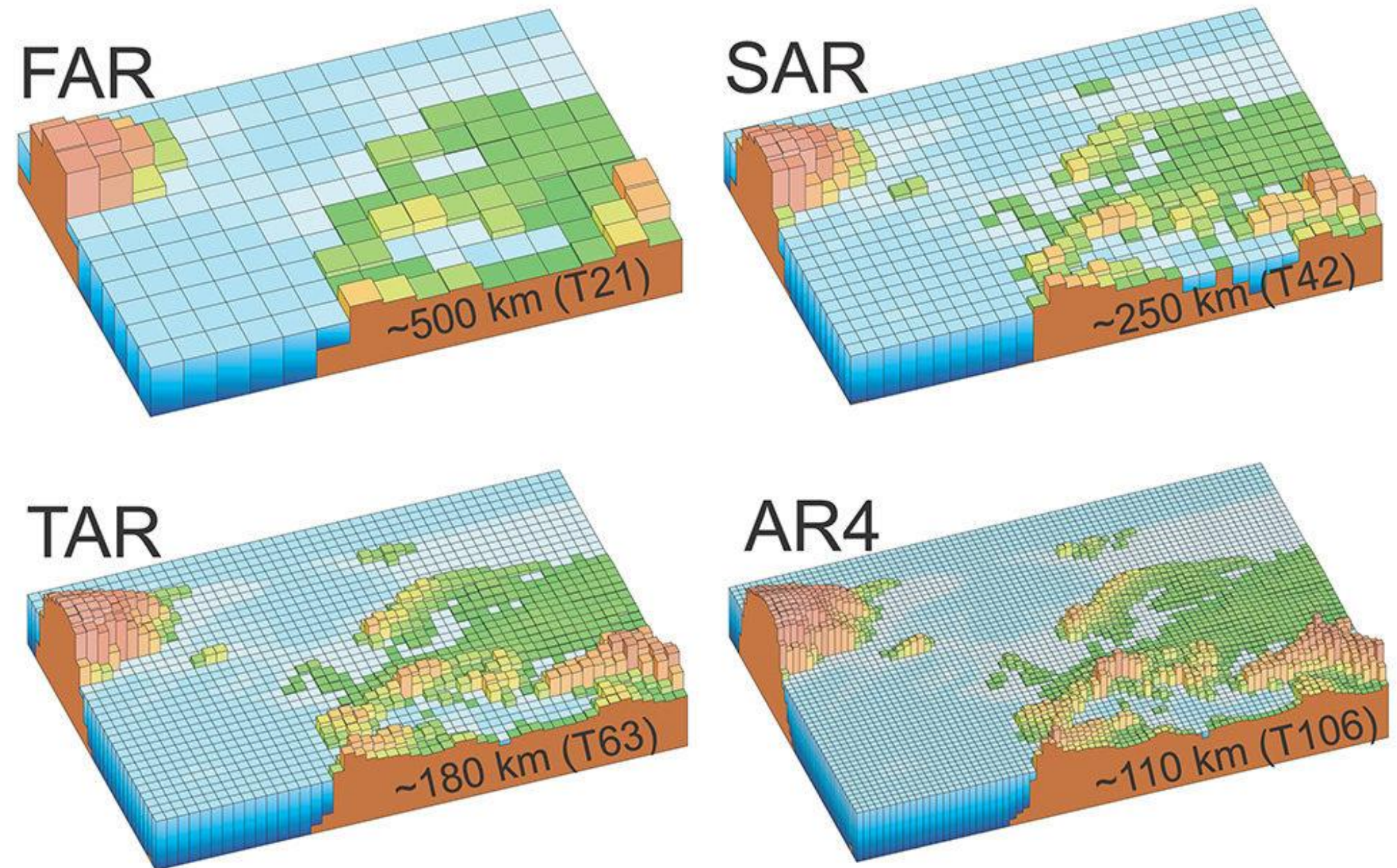
Global Climate Models



The equations derived from these laws are so complex that they must be solved numerically. As a consequence, climate models provide a solution which is discrete in space and time, meaning that the results obtained represent averages over regions, whose size depends on **model resolution**, and **for specific times**.

[Talk by Gavin Schmidt:](https://www.youtube.com/watch?v=JrJJxn-gCdo)

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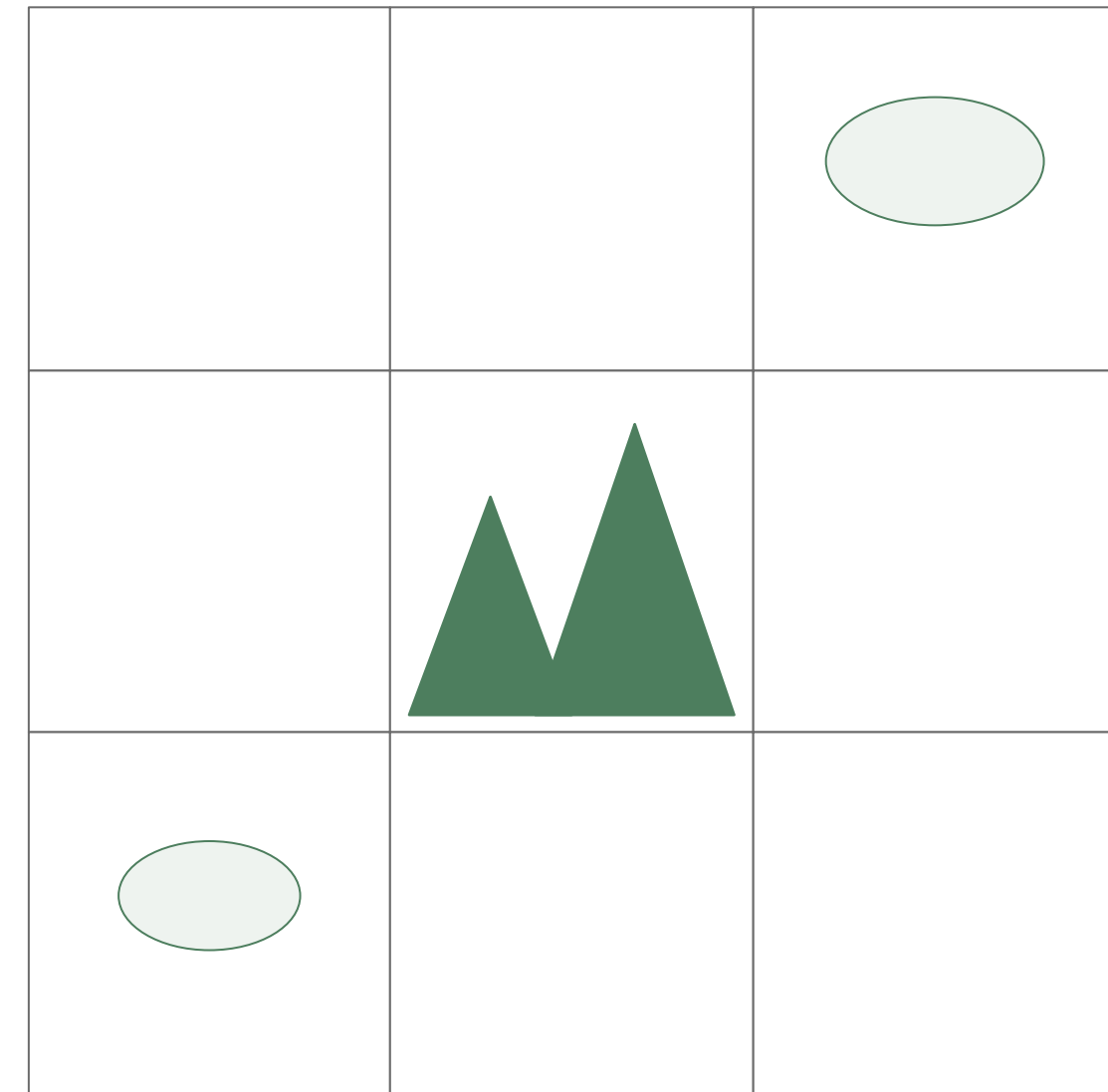
Increasing spatial resolution of climate models used through the first four IPCC assessment reports: first ("FAR") published in 1990, second ("SAR") in 1995, third ("TAR") in 2001 and fourth ("AR4") in 2007. (Note, there is also a fifth report, which was completed in 2014). Source: [IPCC AR4, Fig 1.2](#)

Parameterization and sub-scale grid processes

Parameterization → approximating sub-grid processes using simplified equations.

Some common parameterization processes include:

- Cloud microphysics: droplet information, ice crystal growth and how clouds interact with radiation.
- Convection: Thunderstorms, tropical cyclones and vertical air movement within a grid cell.
- Boundary layer **turbulence**: Surface wind stress and heat fluxes near the ground.
- **Ocean eddies**: Mesoscale circulation features (roughly 10-100 km) that transport heat and nutrients.

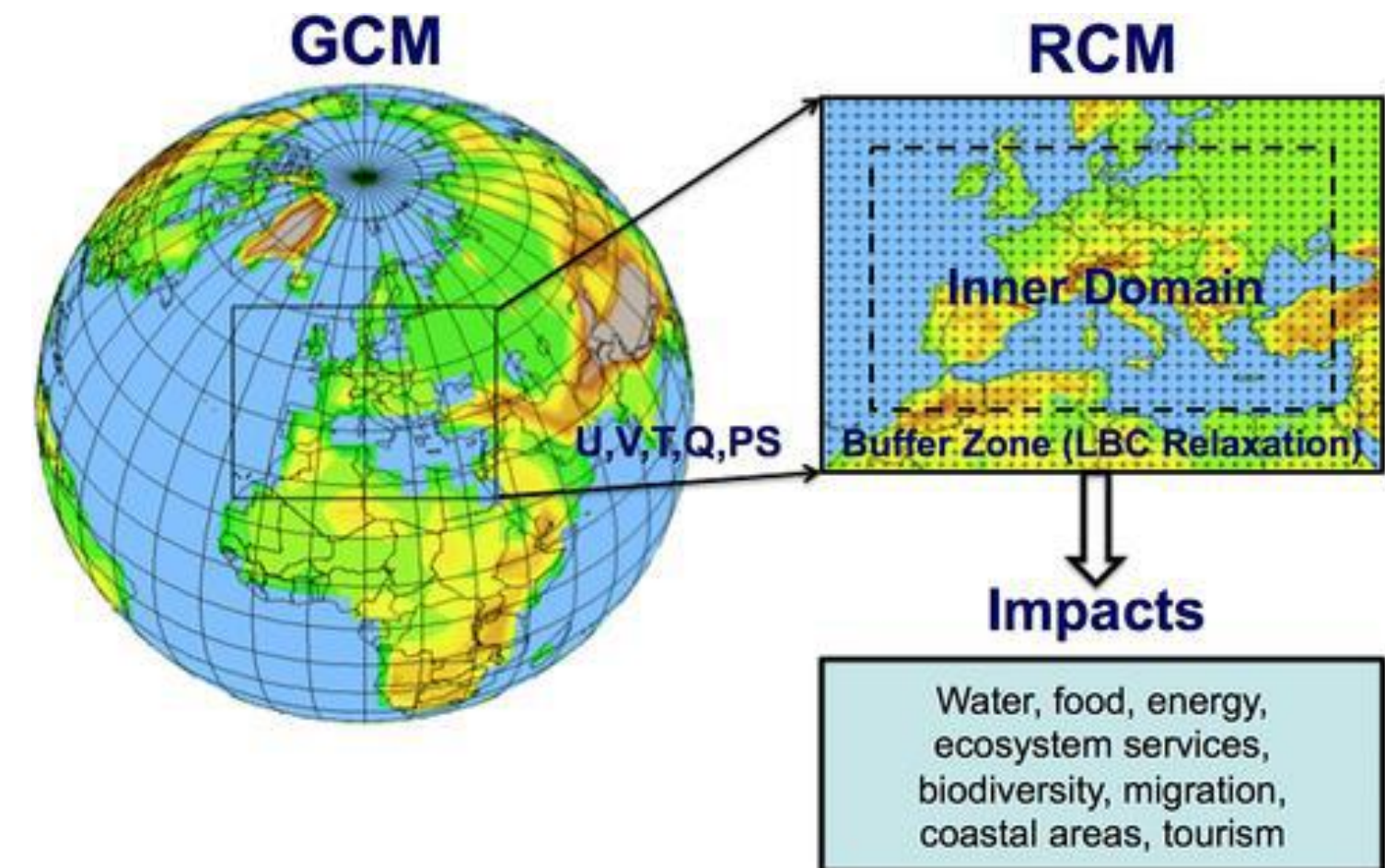


One 100 km cell sees the Karakoram as a single average height. K2 and the Hunza valley get the same number.

*Look for **Karakoram Anomaly***

Regional Climate Models (RCMs) to solve the complexity of GCMs

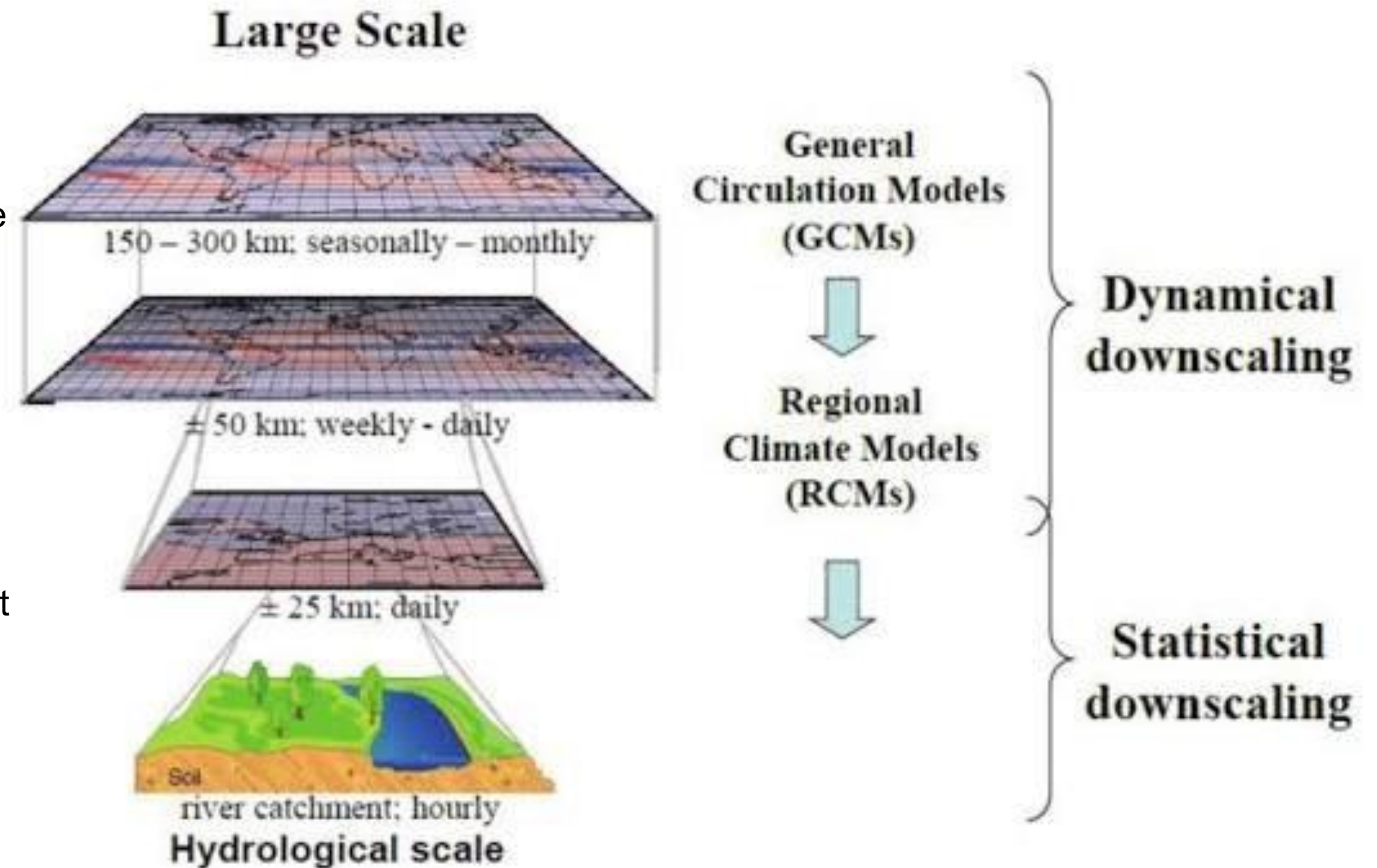
- A global model at 100 km sees as one average height and the plain as a handful of boxes. That is too coarse for decisions, and it means features with a strong local character such as mountain precipitation, coastal circulation, monsoon rainfall over complex terrain are either smeared out or missing.
- An RCM is a full numerical climate model run over a limited area, at 10 to 25 km rather than 100 km, taking its lateral boundary conditions from a GCM or from reanalysis. Winds, temperature, humidity and pressure flow in at the edges of the domain; inside it, the model resolves topography and surface processes the global model could not. This is called **one-way nesting**, because information travels from the global model into the regional one and never back out.
- It adds sub-grid-scale detail to whatever it is given. If the global model has the monsoon in the wrong place, the regional model will place it wrongly in greater detail.



Statistical and Dynamical downscaling

Dynamical downscaling runs actual physics on a finer grid. The regional model solves the same conservation equations as the GCM, but over a smaller domain where it can afford the resolution. It produces physically consistent fields rainfall, temperature, wind and humidity that agree with each other which matters if the output will drive a hydrological or crop model.

Statistical downscaling instead learns a relationship. First an empirical link is established between large-scale predictors, such as sea level pressure, and a local variable, such as station rainfall. That relationship is then applied to GCM output to generate local projections. It is computationally cheap and works well where long observational records exist, but it assumes the historical relationship still holds in a changed climate.



Calibration & Validation

- The factor most often **tuned** for – in 70% of cases – is the radiation balance at the top of the atmosphere.
- Scientists test, or “**validate**”, their models by comparing them against real-world observations. This might include, for example, comparing the model projections against actual global surface temperatures over the past century.
- Scientists do not “**tell**” their models how the climate has changed in the past they do not feed in historical temperature readings.
- Instead, they feed in information on past climate forcings, and the models generate a “**hindcast**” of historical conditions.

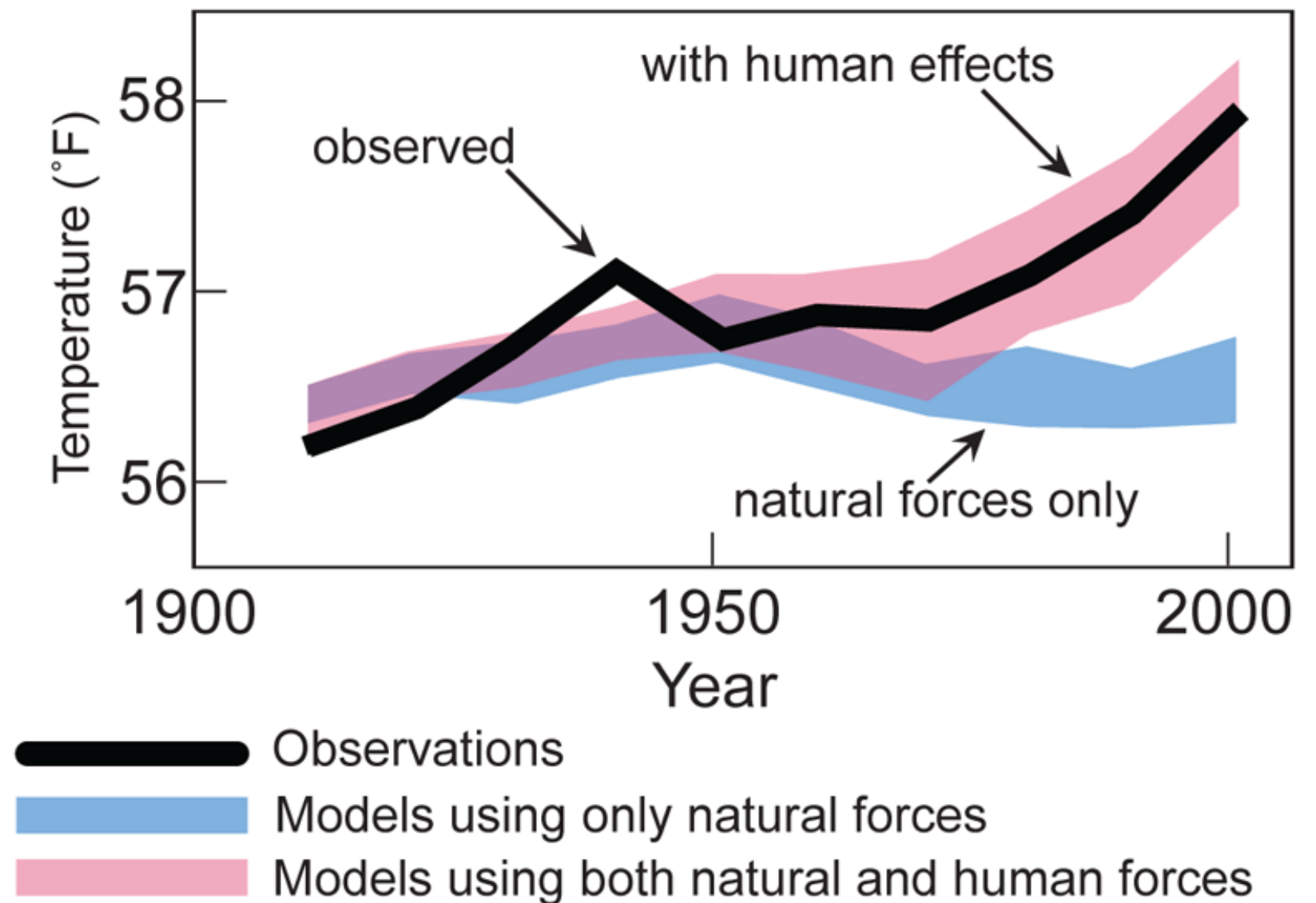


Figure from the IPCC's Fourth Assessment Report ([Hegerl et al 2007](#)).

Typical GCM inputs and boundary conditions

Past forcings

Greenhouse gases from ice cores

Volcanic aerosols

Solar output

Orbital cycles

Land use change

Future forcings*

Emission scenarios: RCPs, SSPs

Aerosol pathways

Land use pathways

All come from socio-economic storylines, not physics

Boundary conditions

Topography

Ocean bathymetry

Initial state, then spin-up until the model settles

Early years discarded

Temperature & Humidity are main **outputs** of GCMs

- temperatures and humidity of different atmospheric layers from the surface to the upper stratosphere.
- Climate models also produce estimates of ocean temperatures, salinity and pH from the surface to the seafloor as well as snowfall, rainfall, snow cover and the extent of glaciers, ice sheets and sea ice.
- They also give information about wind speed, strength and direction, as well as climate features, such as the jet stream and ocean currents.
- “**Climate sensitivity**” can also be modelled (i.e. the warming expected when the concentration of carbon dioxide in the atmosphere reaches twice the amount it was in preindustrial times).

~3 °C

Equilibrium climate sensitivity: warming per doubling of CO₂.

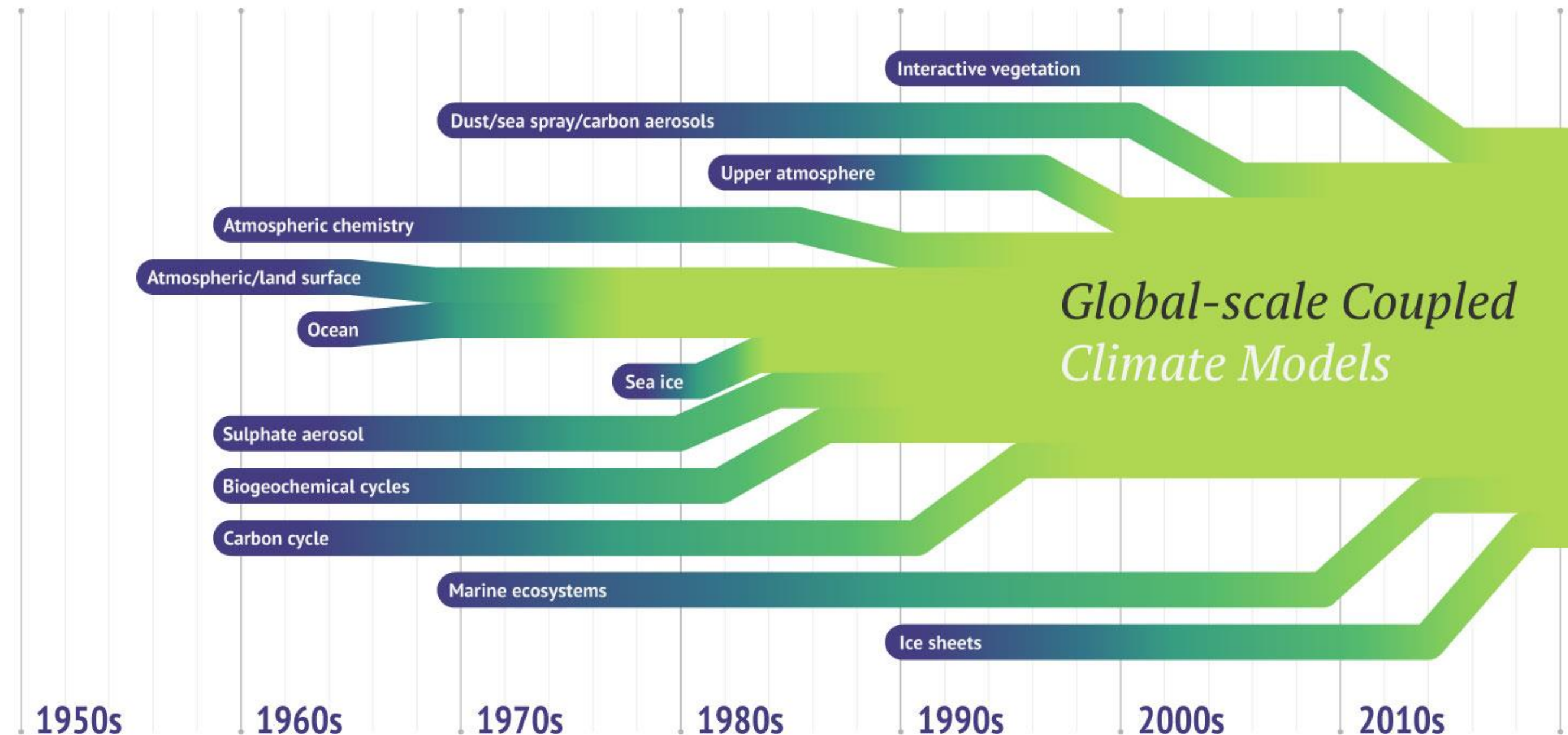
Climate models are not a perfect representation of the Earth's climate

- As the climate is inherently chaotic, it is impossible to simulate with 100% accuracy, yet models do a pretty good job at getting the climate right.
- The accuracy of projections made by models is also dependent on the quality of the forecasts that go into them.
- Similarly, there are aspects of the future that would be so rare in Earth's history that they're extremely difficult to make projections for.
- One example is that ice sheets could destabilize as they melt, accelerating expected global sea level rise.



Climate models

For decades scientists have been using **mathematical models** to help us learn more about the Earth's climate. Known as climate models, they are driven by the fundamental physics of the atmosphere and oceans, and the cycling of chemicals between living things and their environment. Over time they have increased in complexity, as separate components have merged to form **coupled systems**.



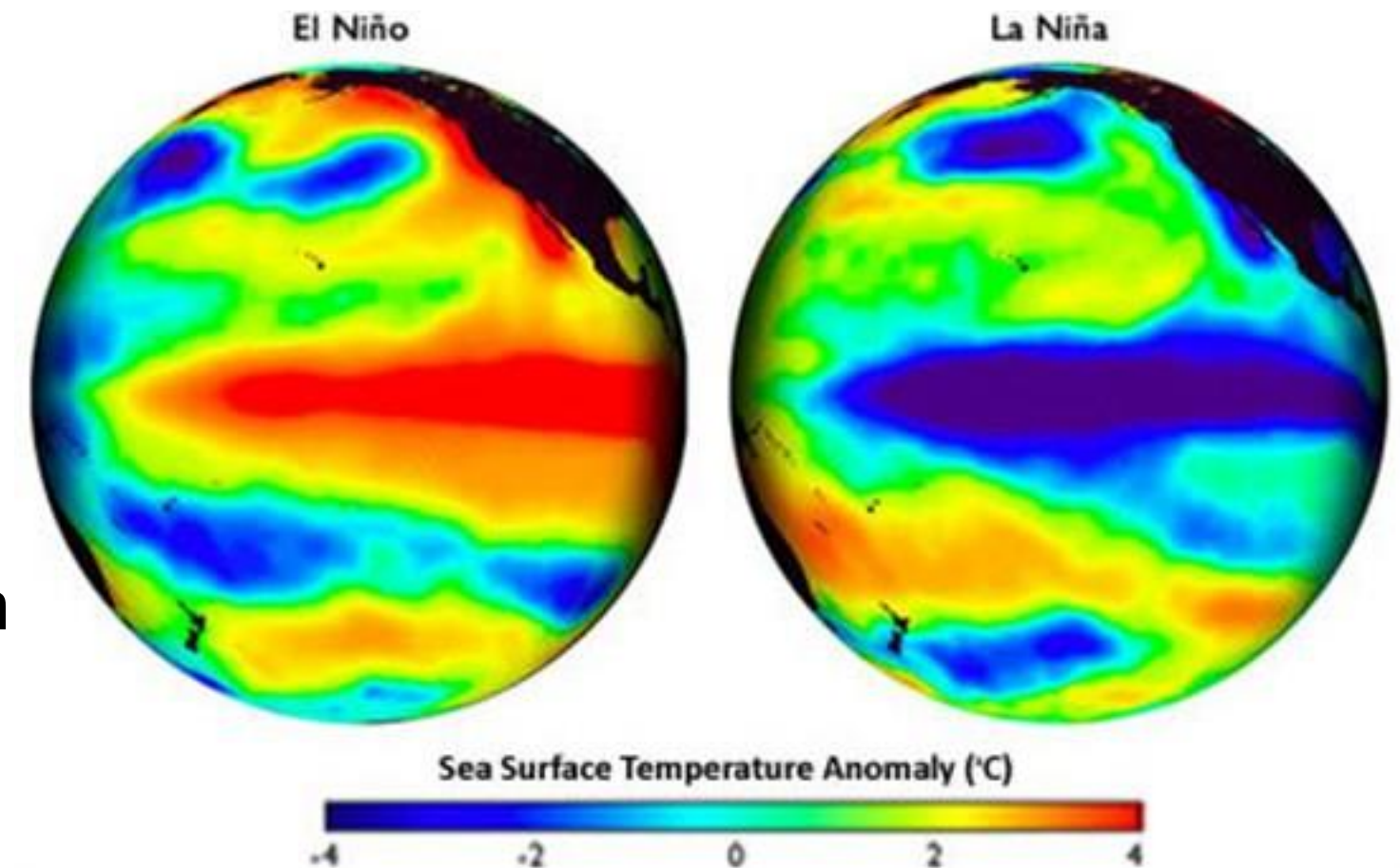
Note: There were some very simplified models before the dates mentioned.

ESMs: Coupled **Atmosphere-Ocean** Models for Climate Simulations

Coupled models simulate atmosphere and ocean interactions together rather than treating each system in isolation.

This coupling is what allows models to reproduce important modes of climate variability. **El Niño-Southern Oscillation (ENSO)**, for example, involves a feedback loop between tropical Pacific sea surface temperatures and atmospheric circulation patterns.

Coupled models also represent **ocean heat uptake**, which is critical for understanding Earth's energy balance. The ocean absorbs and redistributes enormous amounts of heat, and this process helps explain why surface warming doesn't always track emissions in a straight line.



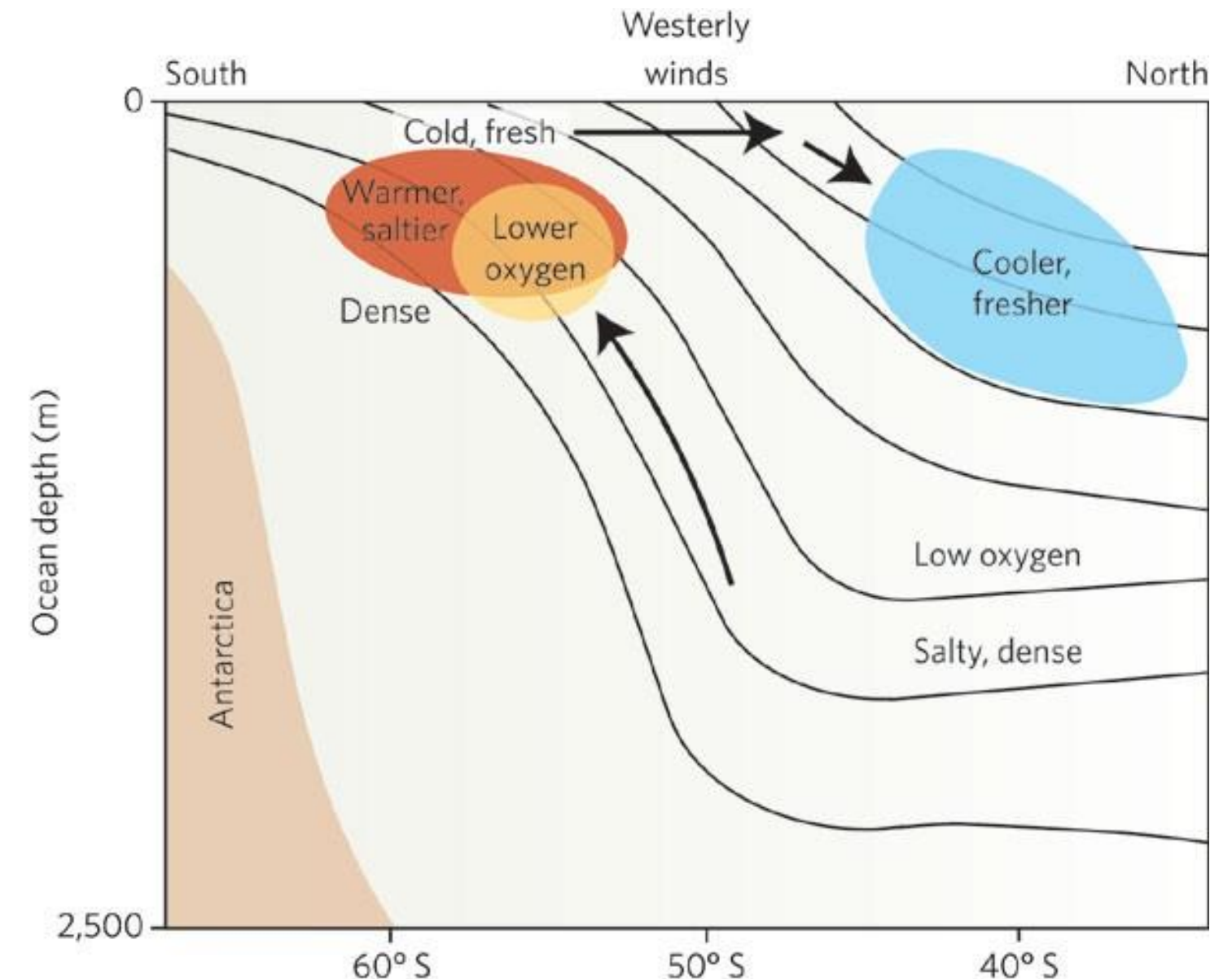
The apparent slowdown in surface warming during the early 2000s, for instance, was partly linked to increased heat uptake in the deep ocean.

ESMs: Ocean Circulation and Climate Impacts

Beyond surface interactions, coupled models simulate how changes in ocean circulation ripple through the climate system:

- The **Atlantic Meridional Overturning Circulation (AMOC)** carries warm water northward and influences North Atlantic and European climate. Models project that AMOC could weaken under continued warming, with significant regional consequences.
- **Sea ice feedbacks** connect ocean circulation to polar climate. As sea ice melts, it exposes darker ocean water that absorbs more solar radiation, amplifying warming. This is a key driver of **Arctic amplification**, the observation that the Arctic is warming roughly two to four times faster than the global average.

The ocean's role as a **carbon sink** is also captured in coupled models. Oceans have absorbed approximately 30% of anthropogenic CO₂ emissions but warming and acidification are reducing this uptake capacity over time.



Toggweiler, J., Russell, J. Ocean circulation in a warming climate. *Nature* **451**, 286–288 (2008). <https://doi.org/10.1038/nature06590>

Biogeochemical Cycles in ESMs: Carbon Cycle Integration

The carbon cycle is what most clearly separates an ESM from a GCM. ESMs simulate CO₂ exchange between atmosphere, land biosphere and ocean — photosynthesis, respiration, ocean dissolution.

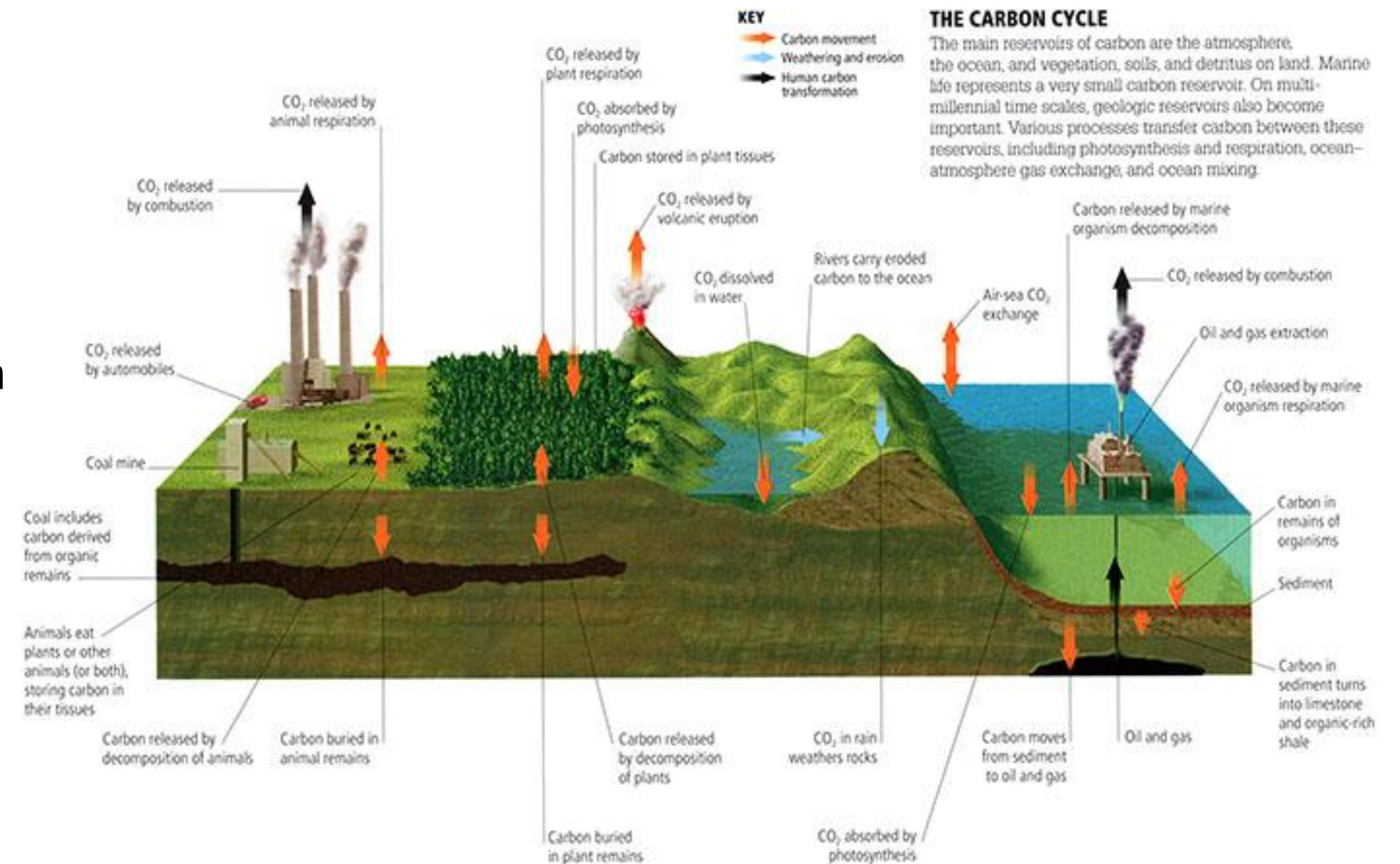
This matters because it enables climate-carbon feedbacks.

A GCM is told the CO₂ concentration. An ESM computes warming changes the carbon cycle, which changes CO₂, which changes warming.

Example:

- **Permafrost thaw** — warming Arctic soils release stored carbon as CO₂ and methane.
- **Weakening ocean sink** — warmer water dissolves less CO₂, and acidification disrupts the organisms that sequester carbon.

This is why carbon budgets come from ESMs, not GCMs.

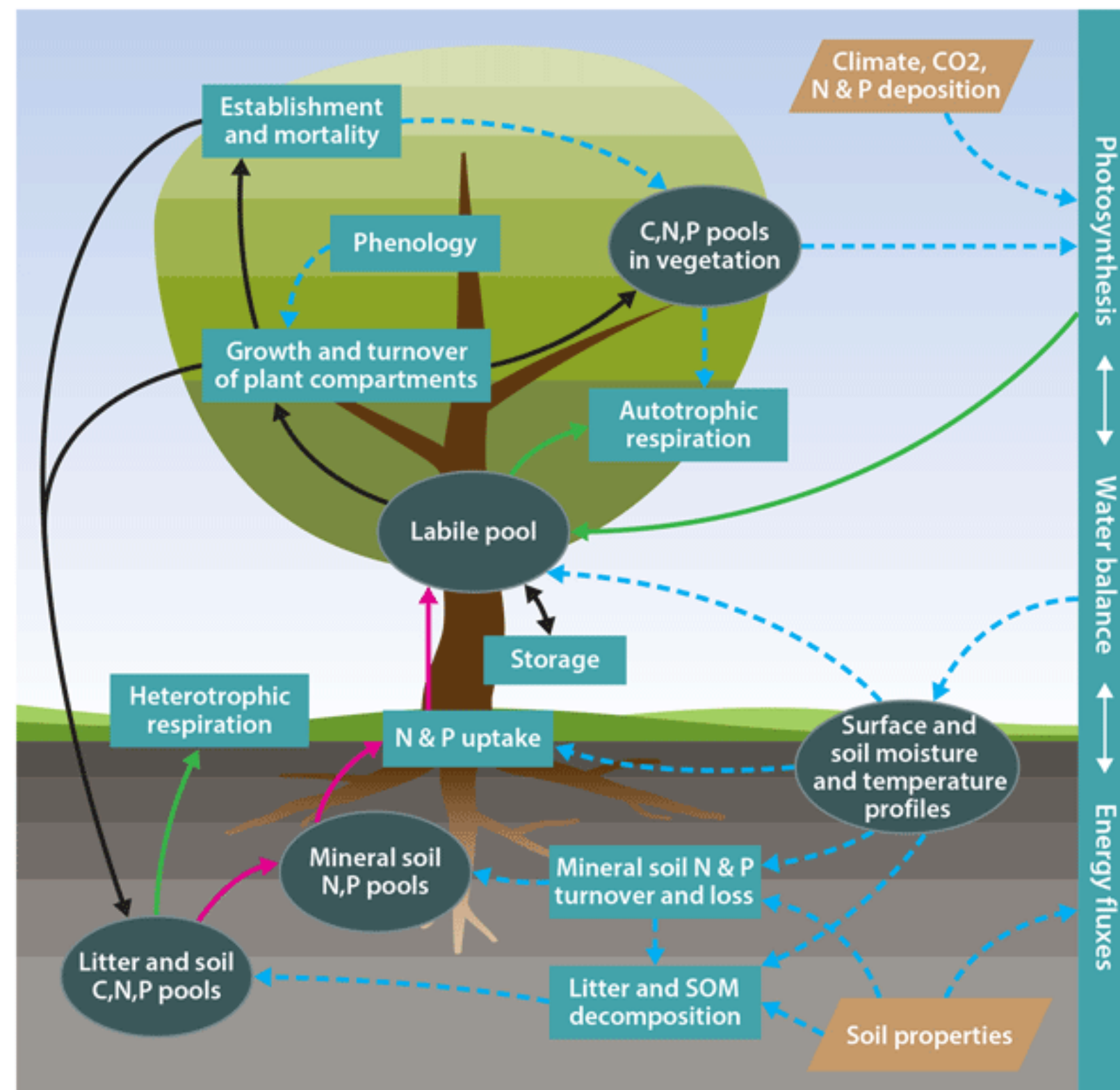


Biogeochemical Cycles in ESMs: Nutrient Cycles & Ecosystem Dynamics

Carbon uptake by ecosystems doesn't happen in a vacuum. Plants need nitrogen and phosphorus to grow, and these nutrients are often in limited supply:

- **Nitrogen limitation** is the primary constraint on plant growth in boreal forests and many temperate ecosystems. Even if CO₂ levels rise (which can stimulate photosynthesis), plants can't take full advantage without sufficient nitrogen.
- **Phosphorus limitation** dominates in tropical rainforests, where ancient, weathered soils have been depleted of this nutrient over millennia.

ESMs also simulate how **land use changes** affect greenhouse gas budgets. Deforestation releases stored carbon, cropland expansion alters surface albedo and moisture fluxes, and fertilizer application increases nitrous oxide (N₂O) emissions, a potent greenhouse gas. These interactions create feedback loops between vegetation growth, nutrient availability, carbon uptake, and climate that only ESMs can capture.



Thum, T., et al.: A new model of the coupled carbon, nitrogen, and phosphorus cycles in the terrestrial biosphere (QUINCY v1.0; revision 1996), Geosci. Model Dev.,

Typical Outputs of GCMs & ESMs: Atmosphere

Variable	nomenclature	Why it matters
Near-surface temperature	tas	Headline variable; models agree most here
Daily maximum temperature	tasmax	Heat thresholds for labour, schooling, mortality
Precipitation	pr	Monsoon totals, reservoir inflow, rainfed agriculture
Sea level pressure	psl	Monsoon onset and circulation
Specific humidity	huss	With temperature gives wet-bulb, the survivability limit
Downward solar radiation	rsds	Solar PV yield, energy planning
Winds	ua, va	Wind resource; westerlies bringing Karakoram snow
Cloud cover	clt	Largest source of inter-model disagreement

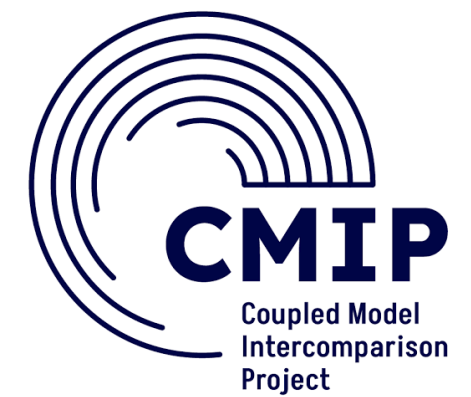
Typical Outputs of GCMs & ESMs: Ocean and Ice

Variable	nomenclature	Why it matters
Sea surface temperature	tos	for example Arabian Sea warming, cyclones, monsoon moisture
Sea level	zos, zostoga	saline intrusions at delta
AMOC strength	msftmz	Headline tipping-point indicator
Sea ice	siconc	Arctic amplification
Ocean pH	ph	Acidification, independent of temperature

Typical Outputs of GCMs & ESMs: Land

Variable	nomenclature	Why it matters
Soil moisture	mrso	Agricultural drought, sowing dates, irrigation demand
Runoff	mrro	Input to any hydrological model
Snow water equivalent	snw	Snow melting
Latent heat flux	hfls	Evapotranspiration, crop water requirement
Soil moisture	mrso	Agricultural drought, sowing dates, irrigation demand

Coupled Model Intercomparison Project (CMIP)



- The Coupled Model Intercomparison Project is a standard experimental framework for coupled atmosphere-ocean models. Every centre runs its model with the same forcings, the same experiment design, and publishes output in the same standardised format. This is what allows disagreement between models to be attributed to the models themselves rather than to how they were set up.
- CMIP is run by the World Climate Research Programme and is independent of the IPCC, though its output underpins the assessment reports. A new phase has appeared roughly every five to six years since 1995.
- Its objectives have barely changed: test how realistically models reproduce the recent past, project climate on both near-term and long-term horizons, and explain why models differ from one another.



<https://wcrp-cmip.org/map/>